

Deterministic Choice Probabilities and Share Inversion for Factor Probit with Many Alternatives

Peter Cotton*

August 2026 — working draft, revised

Abstract

Multinomial probit choice probabilities are high-dimensional Gaussian integrals with no closed form; the standard tool, the GHK simulator, computes them one alternative at a time by smoothed importance sampling. For the factor-structured case $\Sigma = VV^\top + \text{diag}(D)$ we describe a deterministic method that computes *all* N probabilities together: conditionally on k Gaussian factors the alternatives are independent, and a lattice survival-product identity yields the entire share vector in $O(QNL)$ for Q quadrature nodes and L lattice points — deterministic quadrature at low factor rank, reproducible fixed-seed RQMC at higher rank. In the tested $k = 2$ problems the method had observed maximum discrepancy below 10^{-3} from the stated simulation references, from $N = 5$ to $N = 5000$ (22 seconds at $N = 5000$, where the reference’s own noise is of comparable order), and became faster than our reference GHK implementation near $N = 200$; extrapolating that implementation’s timing and $R^{-1/2}$ accuracy suggests hours, not seconds, at matched error for $N = 5000$. A direct utility simulator at matched wall time reaches errors within a factor of 1.3–3 at these budgets, so at the 10^{-3} level the case for the method is not raw forward speed but error decay, reproducibility, resampling-free derivatives, and inversion. Because the approximation is reproducible and differentiable without resampling, it supports practical market-share inversion (a Jacobi-style diagonal-Newton fixed point; $N = 1000$ in under a minute) and reuse of the conditional field for all N single-removal counterfactuals in $O(QN^2L)$. Boundary experiments are reported as they fell: on dense covariances with three spectral-decay profiles, the rank-8 approximation matched or beat GHK at $R = 10^4$ throughout the tested size, and in a substitution study with all noise families standardized to unit variance, factor structure supplied the larger accuracy increment on high-mass deletions while family and factor increments were comparable on mid-mass deletions, in this Gaussian-adjacent design.

1 Introduction

The multinomial probit (MNP) model computes choice among N alternatives with jointly normal utilities, allowing flexible substitution patterns — the property that plain multinomial logit trades away for tractability. Its probabilities are $(N - 1)$ -dimensional Gaussian orthant integrals. The field’s workhorse for three decades has been the GHK simulator [1, 2, 3]: smoothed

*Numbers in this paper are produced by committed, seeded scripts with correctness anchors run before any comparison (<https://github.com/microprediction/kinetics>, experiments 13–16; an immutable commit will be pinned at submission). The algorithm ships in the `thurstone` package; package–benchmark agreement is itself a reported number. Benchmarks: Apple M4, 16 GB, Python 3.12.9, NumPy 2.4.6, SciPy 1.18.0, single-threaded; timings are single realizations without warm-up medians; peak memory below 2 GB.

importance sampling along a Cholesky factor, one alternative at a time. GHK is accurate and well understood at moderate N ; its structural costs are that error decays as $R^{-1/2}$ in the number of draws, that the full share vector requires a separate computation per alternative, and that each evaluation is a finite-simulation approximation whose characteristics enter derivative-based estimation (Section 2).

This paper concerns the factor-structured case

$$U_i = \mu_i + v_i^\top f + \sqrt{D_i} \varepsilon_i, \quad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0, 1) \text{ i.i.d.}, \quad D_i > 0, \quad (1)$$

i.e. $\Sigma = VV^\top + \text{diag}(D)$ — the form in which applied covariance structure often arrives. Factor dimension reduction for probit is itself classical: one-factor quadrature goes back to Butler and Moffitt [4], and factor-analytic probit to Elrod and Keane [5]. What appears not to have been assembled is the *all-alternatives* computation: conditionally on the factors, a lattice survival-product identity [9] computes every alternative’s one-dimensional integral in a single $O(NL)$ pass — the field survival product is built once and each alternative’s leave-one-out field is recovered by division. A k -dimensional quadrature around that inner loop gives the whole share vector deterministically. The contribution here is that assembly, its inverse (share-to-utility) transform, the reuse of the conditional field for removal counterfactuals, and benchmark plus boundary evidence.

Proposition 1 (Conditional identity). *Under (1), with $g_i(\cdot | f)$ and $F_i(\cdot | f)$ the conditional density and CDF of U_i given f ,*

$$p_i = \mathbb{E}_f \left[\int g_i(x | f) \prod_{j \neq i} F_j(x | f) dx \right], \quad \sum_i p_i = 1, \quad (2)$$

and the product over $j \neq i$, for all i simultaneously, is obtained from $\log F_{\text{field}} = \sum_j \log F_j$ by subtraction.

Proof. Conditional on f the U_j are independent, so $\mathbb{P}(U_i \text{ maximal} | f) = \int g_i \prod_{j \neq i} F_j dx$; average over f . The events partition (ties are null for continuous laws), giving $\sum_i p_i = 1$. $\square$

Proposition 2 (Complexity). *With Q factor nodes and L lattice points, the forward share vector costs $O(QNL)$; the full single-removal ensemble $\{P(j \text{ chosen} | i \text{ removed})\}_{i,j}$ costs $O(QN^2L)$ arithmetic and $O(N^2)$ storage, reusing one conditional field construction.*

Proof. Per node: the field log-product costs $O(NL)$, each alternative’s integral $O(L)$ after subtraction, giving $O(NL)$; the removal ensemble evaluates N survivor integrals for each of N removals, $O(N^2L)$, over the same cached field. $\square$

Proposition 3 (Identification and differentiability). *$p(\mu + c\mathbf{1}) = p(\mu)$, so $J = \partial p / \partial \mu$ has $\mathbf{1}$ in its null space and μ is identified only up to location. With B an $N \times (N - 1)$ basis of $\mathbf{1}^\perp$ and $\mu = B\eta$, share inversion and implicit differentiation are taken in reduced coordinates: $d\eta/d\theta = -(B^\top JB)^{-1} B^\top \partial p / \partial \theta$, provided $B^\top JB$ is nonsingular.*

Proof. Invariance holds because a common location shift does not change the argmax; differentiating $p(\mu + c\mathbf{1}) = p(\mu)$ in c gives $J\mathbf{1} = 0$. Differentiating $p(B\eta(\theta), \theta) = \hat{p}$ in θ and projecting with $B^\top$ gives the reduced formula under the stated regularity. $\square$

Remark 1. For the Gaussian model with $D_i > 0$, $p(\mu) = \nabla_\mu G(\mu)$ for the social-surplus function $G(\mu) = \mathbb{E} \max_i \{\mu_i + \xi_i\}$; strict convexity of G on $\mathbf{1}^\perp$ would give injectivity of the share map modulo location and, with a coercivity argument, existence and uniqueness of the inverse for every strictly positive target. We use this only as motivation here; a proof is future work.

Proposition 4 (Jacobian–vector products in $O(QNL)$). *For a direction h , let $A(x, f) = \sum_j h_j \partial_{a_j} \log S_j(x | f)$ in the reflected (min-wins) coordinates a . Then*

$$(Jh)_i = \mathbb{E}_f \int R_i [h_i \partial_{a_i} g_i + g_i (A - h_i \partial_{a_i} \log S_i)] dx, \quad R_i = \prod_{j \neq i} S_j,$$

where the field sum A is formed once per node and lattice location, so all N components cost $O(NL)$ per node. (Verified against central finite differences to 8×10^{-9} ; implemented and tested in the repository.) This enables Krylov solves of the reduced system in Proposition 3 without forming J densely at $O(QN^2L)$.

The factors need not be Gaussian: (2) requires only conditional independence and an integrable factor law, so Student, mixture, or empirical factor distributions are admissible with appropriate nodes, and the base densities may be arbitrary and alternative-specific. One caution for asymmetric laws: the implementation works in the reflected (min-wins) race, and $-U$ has the same distribution as the reflected model only when the factor and base laws are symmetric; for asymmetric laws the reflected distributions must be used explicitly.

Numerical method, stated completely. The benchmarks use $L = 1501$ lattice points on the adaptive interval $[\min_{i,q} m_i(f_q) - 8 \max_i \sqrt{D_i}, \max_{i,q} m_i(f_q) + 8 \max_i \sqrt{D_i}]$, where $q = 1, \dots, Q$ indexes the retained factor nodes. Factor integration: the GHK benchmark (experiment 13) uses product Gauss–Hermite of order 15/11/9 for $k = 2/3/4$; the boundary experiment (experiment 14) uses order 15 per dimension for all $k \leq 4$, which after pruning weights below 10^{-7} of the maximum leaves $Q = 145, 1317$, and 10929 nodes for $k = 2, 3, 4$ — making the exponential growth of Q with rank explicit; scrambled-Sobol nodes (2^{13} points, fixed seed) are used for $k > 4$: reproducible fixed-seed randomized QMC, disclosed as such. Survival floors sit at 10^{-300} and the quadrature output is normalized by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2); statements about derivatives refer to derivatives of this approximation, whose fidelity to exact MNP derivatives is governed by L and Q .

Inversion. Given target shares $\hat{p}$ and (V, D) , the inverse transform finds mean-zero μ with $p(\mu) = \hat{p}$ by a damped Jacobi-style diagonal-Newton iteration: all coordinates updated simultaneously using own-location slopes of the unnormalized lattice map (available from the same pass), the field refrozen each iteration, a warm start from the independent-field inverse using total variances, and a tail-aware tolerance (alternatives with target shares below $10^{-4}/N$ are matched best-effort; their locations are weakly identified — itself a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear). Convergence in a handful of iterations is an empirical observation on the tested problems, not a theorem. The design follows the original transform’s inversion [9] and the calibration practice of [14].

2 Benchmark against GHK

Anchors first. On the binary case, where MNP has a closed form, the lattice transform agrees to 3×10^{-16} and our GHK implementation — validated before being compared against — to 2×10^{-14} at $R = 2 \times 10^5$ (an easy anchor for both methods, useful for code validation rather than as evidence about the hard regime). At $N = 5$ both methods sit within the noise of a 10^7 -draw Monte Carlo reference, and the shipped `thurstone.FactorRace` agrees with the benchmark kernel to 1.1×10^{-5} .

Error metric: throughout, $\max_i |\hat{p}_i - p_i|$ against a simulation reference. A maximum over N coordinates tends to grow with N at fixed per-coordinate accuracy, and the reference’s own maximum-coordinate noise behaves likewise; mean-absolute and total-variation summaries are committed alongside (experiment 16): at $N = 1000$ the lattice’s mean absolute error is 6×10^{-6} and total variation 3.1×10^{-3} against a fresh 2×10^6 -draw reference.

N	lattice transform	GHK ($R = 1000$)
5	21 ms, err 4.1×10^{-4}	1 ms, err 3.1×10^{-3}
20	95 ms, err 3.8×10^{-4}	12 ms, err 4.4×10^{-3}
50	257 ms, err 2.1×10^{-4}	77 ms, err 7.7×10^{-3}
200	930 ms, err 2.8×10^{-4}	1333 ms, err 6.7×10^{-3}
1000	4.5 s, err $8.0 \times 10^{-4} \dagger$	—
5000	22.6 s, err $9.0 \times 10^{-4} \dagger$	—

Table 1: Full share vector, $k = 2$ factors, one problem per size (utility spread 1.0 through $N = 200$, 1.5 above). Reference: 2×10^6 MC draws through $N = 200$; 5×10^5 draws at $N = 1000$ and 5000 ($\dagger$: there the reference’s own maximum-coordinate noise is of comparable order, so these entries bound the lattice error only to that order). A replication study at common spread 1.0 (experiment 16; ten problems per size at $N \leq 200$, three at $N = 1000$, twin independent 2×10^6 -draw references per problem) gives worst-case lattice error $2.9\text{--}3.6 \times 10^{-4}$ at every size, at or below the references’ own replicate-to-replicate noise. Lattice error stayed below 10^{-3} throughout; GHK cost crossed over by $N \approx 200$.

Matched-accuracy extrapolation (not a measurement). Fitting a power law to the reference implementation’s timings over $N \in [20, 200]$ ($\alpha \approx 2.0$; a lower bound, the asymptotic per-share cost being cubic) and combining with $R^{-1/2}$ error scaling suggests that matching the lattice’s 9×10^{-4} at $N = 5000$ would take this implementation on the order of twelve hours, against 22 seconds. The baseline is a straightforward single-threaded Python GHK: per-alternative Cholesky, plain pseudorandom uniforms, one realization per problem, shares normalized by their sum. It establishes the scaling of the per-alternative simulation approach, not a field-wide impossibility. The strongest cheap baseline is *direct utility simulation* — draw utilities, take argmax, average — which produces the whole share vector per draw and, unlike GHK, does not pay per alternative. We ran it at wall time matched to the lattice call (experiment 16): its maximum-coordinate error was within a factor of 1.3–3 of the lattice’s at every tested N (7.2×10^{-4} versus 4.9×10^{-4} at $N = 1000$, 0.6M draws in the matched 4.4 seconds). At the 10^{-3} accuracy level, then, raw forward speed does not by itself separate the methods; what does is error decay ($R^{-1/2}$ against spectral-in- L , so each added decimal digit costs 100× in draws against a linear lattice refinement), reproducibility, resampling-free derivatives, and the

inversion and removal machinery below. Remaining stronger baselines — QMC-GHK, seed-replicated uncertainty bands, minimax tilting [8] — are queued in the repository.

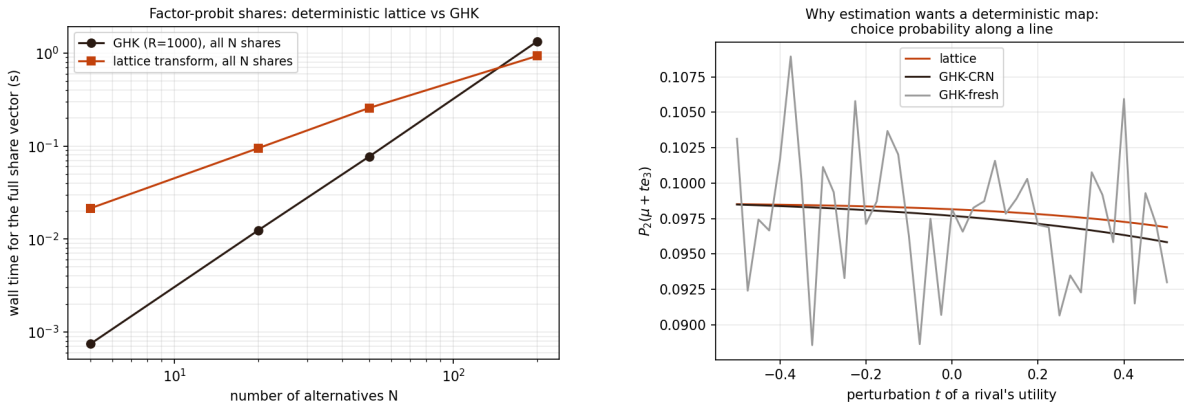


Figure 1: Left: wall time for the full share vector (log–log). Right: a choice probability along a line in utility space. Fresh-draw GHK shows run-to-run simulation noise (roughness statistic — maximum second difference over dt^2 — 39.3 vs. 0.01); common-random-number GHK is a smooth deterministic approximation conditional on its draw set, and its curve differs visibly from the lattice curve — a discrepancy between two approximations whose decomposition requires an independent high-accuracy reference, queued. All three are approximations.

Derivatives, stated correctly. For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of μ whose derivatives are evaluated without resampling; analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws [2, 3]. The practical distinction is the character and controllability of the approximation error — quadrature resolution versus draw count — not the possibility of differentiation.

Share inversion. At $N = 1000$, $k = 2$, with targets from 5×10^6 MC draws floored at 10^{-7} and renormalized: the inversion completes in 59 seconds. The iteration solves its own equations to 6.3×10^{-9} — a statement about the solver, not statistical accuracy, since the target carries sampling noise $\sim 2 \times 10^{-4}$ — and recovers the utilities of identified alternatives (target share above 3×10^{-4}) to maximum error 0.014. A replication study (experiment 16) repeats the inversion on a second $N = 1000$ problem against three independent 5×10^6 -draw targets: 58–61 seconds each, the same 162 alternatives identified each time (98.1% of share mass), recovery error maximum 0.018–0.025 and median 0.0026 across replicates, with the recovery-versus-share curve committed alongside. We are not aware of probit share inversion at this scale in current simulation-based practice.

Removal ensemble. All 200 single-removal share vectors from one conditional field construction in $O(QN^2L)$: $3.7\times$ faster than an extrapolation from direct recomputations, with ten evenly spaced removals checked directly (maximum discrepancy 5.6×10^{-17}).

3 Boundary experiments

General covariance through the factor approximation. The method requires $\Sigma \approx VV^\top + D$; GHK does not. We generated one random correlation matrix at each spectral decay $\lambda_m \propto m^{-\gamma}$, $\gamma \in \{0.5, 1.5, 3\}$, at $N = 50$ (single realizations; replication over matrices and utility vectors is queued), fit factors by iterated principal factor analysis, and measured share error against an 8×10^6 -draw reference, with GHK at the exact Σ for comparison. The plotted quantity is the *rank- k approximation error of the stated fitting and integration procedure* — covariance-fit error plus factor-integration and lattice error — not a minimum over factor models. Findings as they fell: error decays fastest for near-low-rank truth ($\gamma = 3$: 5.6×10^{-2} at $k = 1$ to 1.2×10^{-3} at $k = 8$) and slowest at intermediate decay ($\gamma = 1.5$: 2.4×10^{-2} to 2.6×10^{-3}); at every tested decay the $k = 8$ result (fixed-seed RQMC nodes) matched or beat GHK at $R = 10^4$ (3.2 – 4.1×10^{-3}). We expected a clear GHK-wins regime at this size and did not find one; where accuracy demands exceed the achievable rank- k error — intermediate spectra with k capped, or short-range correlation at large N — GHK’s arbitrary- Σ generality is the right tool. Further practical boundaries (small $D_{\min}/D_{\max}$, wide utility ranges, near-perfect correlation, extreme shares) stress the uniform lattice and are queued as resolution sweeps in L and Q .

Substitution fidelity, with noise families standardized. Why want probit-style flexibility at scale? Ground truth misspecified for every candidate: $t(5)$ -distributed factors and skew-normal idiosyncratic noise, *all noise families standardized to mean zero and unit variance*. (An earlier version of this experiment confounded noise family with noise variance; the standardized rerun is reported here, and the ordering survived.) Candidates — plain logit, factor mixed logit (unit-variance Gumbel base), factor probit — are calibrated to identical full-menu shares with the true loadings supplied (calibration residuals 4.9×10^{-11} and 1.1×10^{-9} on menu shares), and scored on deletion counterfactuals against fresh simulation truths. Blocks whose removed share sits at the Monte Carlo noise floor are uninformative (all models tie). Of 24 deletion blocks, one falls in the high-mass stratum and four in the mid-mass stratum — small strata, so the numbers below are indicative, not estimates with useful standard errors. The misallocated fraction of redistributed mass is:

	deleted mass > 10%	deleted mass 2–10%
plain logit (IIA)	15.9%	25.7%
factor mixed logit	7.7%	17.1%
factor probit	2.8%	7.8%

The precise reading: in the high-mass stratum (a single block), factor structure supplies the larger increment ($15.9 \rightarrow 7.7$, i.e. 8.2 points, versus $7.7 \rightarrow 2.8$, 4.9 points for the idiosyncratic family); in the mid-mass stratum (four blocks) the two increments are comparable, with the family increment slightly larger (8.6 versus 9.3 points). Caveats: the truth’s skew-normal noise is Gaussian-adjacent (favoring probit), and this is a single design and seed with small strata — replication across skewness and tail designs is queued before any general claim.

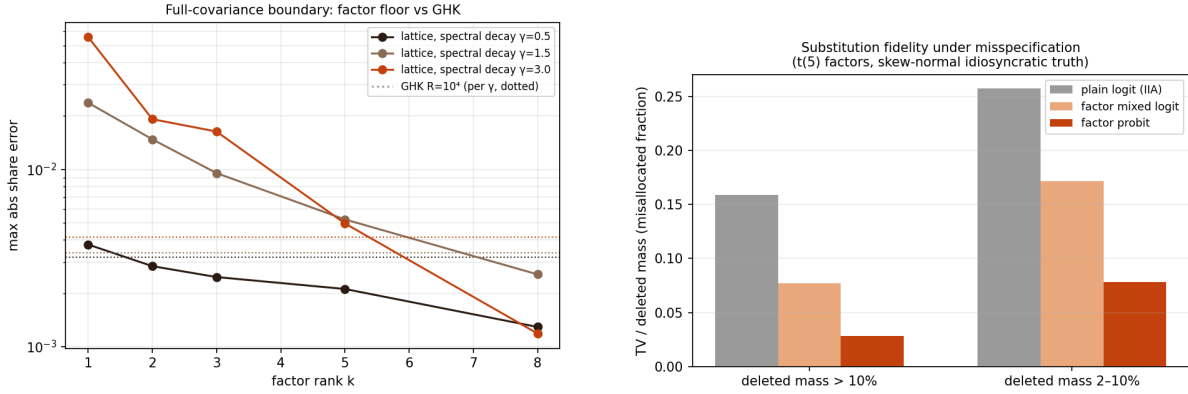


Figure 2: Left: rank- k approximation error under spectral decay γ (dotted: GHK at $R = 10^4$, per γ). Right: substitution fidelity with standardized noise families, stratified by deleted share mass.

4 Related work

GHK and simulated likelihood are standard [1, 2, 3]; analytic-approximation lines include MACML [6]; modern Gaussian-probability methods include minimax tilting [8]. Factor dimension reduction for probit is classical: Butler–Moffitt one-factor quadrature [4], Elrod–Keane factor-analytic probit [5], and the probit probability approximations surveyed in [7]. The lattice survival-product identity and its inverse transform are from [9]. The novelty claimed here is the all-alternatives assembly with inversion and removal-ensemble reuse — not dimensional reduction, which is old. Share inversion and its acceleration are central to demand estimation [10, 11]; automatic differentiation of logit-based BLP is recent [12]; a 2026 preprint approximates correlated-choice probabilities with equivariant neural networks [13], evidence of current demand for this computation. We are unaware of an automatically differentiable, large-choice-set factor-probit demand implementation built on deterministic share inversion; that precise object is what Section 5 proposes.

5 Discussion

Estimation is the natural next step: with the reduced coordinates of Proposition 3, gradients of an outer likelihood or GMM objective pass through the share-inversion fixed point by implicit differentiation, a candidate route to differentiable probit demand estimation at large N . Estimating (V, D) rather than supplying them raises the usual identification cautions: utility scale normalization, factor rotation, choice-invariant common loading components, and the fact that only difference covariances are identified. Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities; a *common-scale* Gumbel base with zero loadings reproduces the Luce model exactly (alternative-specific scales break Luce, as they should), yielding correlated-softmax generalizations. Finally, the inversion inherits the original transform’s interpolation structure (per-node location-to-probability curves are cross-correlations, evaluable at all offsets simultaneously), which we have not yet exploited.

References

- [1] J. Geweke, M. Keane, D. Runkle. Alternative computational approaches to inference in the multinomial probit model. *Review of Economics and Statistics* 76 (1994) 609–632.
- [2] V. Hajivassiliou, D. McFadden. The method of simulated scores for the estimation of LDV models. *Econometrica* 66 (1998) 863–896.
- [3] K. Train. *Discrete Choice Methods with Simulation*, 2nd ed. Cambridge, 2009.
- [4] J. S. Butler, R. Moffitt. A computationally efficient quadrature procedure for the one-factor multinomial probit model. *Econometrica* 50 (1982) 761–764.
- [5] T. Elrod, M. P. Keane. A factor-analytic probit model for representing the market structure in panel data. *Journal of Marketing Research* 32 (1995) 1–16.
- [6] C. Bhat. The maximum approximate composite marginal likelihood (MACML) estimation of multinomial probit-based unordered response choice models. *Transportation Research B* 45 (2011) 923–939.
- [7] R. Connors, S. Hess, A. Daly. Analytic approximations for computing probit choice probabilities. *Transportmetrica A* 10 (2014) 119–139.
- [8] Z. Botev. The normal law under linear restrictions: simulation and estimation via minimax tilting. *JRSS-B* 79 (2017) 125–148.
- [9] P. Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM J. Financial Mathematics* 12 (2021) 295–317.
- [10] S. Berry, J. Levinsohn, A. Pakes. Automobile prices in market equilibrium. *Econometrica* 63 (1995) 841–890.
- [11] C. Conlon, J. Gortmaker. Best practices for differentiated products demand estimation with pyblp. *RAND Journal of Economics* 51 (2020) 1108–1161.
- [12] A. Chia. Automatically differentiable random coefficient logistic demand estimation. arXiv:2106.04636 (2021).
- [13] E. Huch, M. Keane. Amortized inference for correlated discrete choice models via equivariant neural networks. arXiv:2603.24705 (2026).
- [14] The `allocation` package: Thurstone-tilt portfolio construction. <https://github.com/microprediction/allocation>.